

```
In [5]: import pandas as pd
        from sklearn import preprocessing
```

```
In [6]: df=pd.read_csv("C:/Users/Dell/Mall_Customers.csv")
```

```
In [7]: df
```

```
Out[7]:
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

```
In [8]: df.describe()
```

```
Out[8]:
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
In [9]: df.min()
```

```
Out[9]: CustomerID      1
        Genre          Female
        Age            18
        Annual Income (k$) 15
        Spending Score (1-100) 1
        dtype: object
```

```
In [10]: df.max()
```

```
Out[10]: CustomerID      200
        Genre          Male
        Age            70
        Annual Income (k$) 137
        Spending Score (1-100) 99
        dtype: object
```

```
In [11]: df.groupby(["Genre"])["Age"].mean()
```

```
Out[11]: Genre
        Female    38.098214
        Male      39.806818
        Name: Age, dtype: float64
```

```
In [12]: df.groupby("Genre")["Age"].median()
```

```
Out[12]: Genre
        Female    35.0
        Male      37.0
        Name: Age, dtype: float64
```

```
In [15]: df.groupby(df["Genre"])["Age"].std()
```

```
Out[15]: Genre
        Female    12.644095
        Male      15.514812
        Name: Age, dtype: float64
```

```
In [17]: df.groupby('Genre')['Annual Income (k$)'].mean()
```

```
Out[17]: Genre
        Female    59.250000
        Male      62.227273
        Name: Annual Income (k$), dtype: float64
```

```
In [18]: df.groupby('Genre').mean()
```

```
Out[18]:
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
Genre				
Female	97.562500	38.098214	59.250000	51.526786
Male	104.238636	39.806818	62.227273	48.511364

```
In [19]: df.groupby(["Genre"]).median()
```

Out[19]:

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
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Genre				
Female	94.5	35.0	60.0	50.0
Male	106.5	37.0	62.5	50.0

In [20]: `df.groupby(['Genre']).std()`

Out[20]:

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
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Genre				
Female	58.276412	12.644095	26.011952	24.11495
Male	57.483830	15.514812	26.638373	27.89677

In [21]: `df`

Out[21]:

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
--	------------	-------	-----	---------------------	------------------------

0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

In [17]: `x=df.drop(axis=1,columns=['Genre'])`

In [23]: `df`

Out[23]:

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

In [24]: x

Out[24]:

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	19	15	39
1	2	21	15	81
2	3	20	16	6
3	4	23	16	77
4	5	31	17	40
...	...	...	...	...
195	196	35	120	79
196	197	45	126	28
197	198	32	126	74
198	199	32	137	18
199	200	30	137	83

200 rows × 4 columns

```
In [26]: enc=preprocessing.OneHotEncoder()  
enc_df=pd.DataFrame(enc.fit_transform(df[['Genre']]).toarray())  
enc_df
```

```
Out[26]:
```

	0	1
0	0.0	1.0
1	0.0	1.0
2	1.0	0.0
3	1.0	0.0
4	1.0	0.0
...	...	...
195	1.0	0.0
196	1.0	0.0
197	0.0	1.0
198	0.0	1.0
199	0.0	1.0

200 rows × 2 columns

```
In [27]: df_encode=x.join(enc_df)
df_encode
```

```
Out[27]:
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)	0	1
0	1	19	15	39	0.0	1.0
1	2	21	15	81	0.0	1.0
2	3	20	16	6	1.0	0.0
3	4	23	16	77	1.0	0.0
4	5	31	17	40	1.0	0.0
...	...	...	...	...	...	...
195	196	35	120	79	1.0	0.0
196	197	45	126	28	1.0	0.0
197	198	32	126	74	0.0	1.0
198	199	32	137	18	0.0	1.0
199	200	30	137	83	0.0	1.0

200 rows × 6 columns

```
In [28]: dfi=pd.read_csv(r"C:\Users\Dell\Iris.csv")
```

```
In [29]: dfi
```

Out[29]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

```
In [33]: set1=(dfi['Species']=='Iris-setosa')
print(dfi[set1].describe())
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	50.00000	50.00000	50.00000	50.00000	50.00000
mean	25.50000	5.00600	3.41800	1.46400	0.24400
std	14.57738	0.35249	0.38102	0.17351	0.10721
min	1.00000	4.30000	2.30000	1.00000	0.10000
25%	13.25000	4.80000	3.12500	1.40000	0.20000
50%	25.50000	5.00000	3.40000	1.50000	0.20000
75%	37.75000	5.20000	3.67500	1.57500	0.30000
max	50.00000	5.80000	4.40000	1.90000	0.60000

```
In [34]: set2=(dfi['Species']=='Iris-virginica')
print(dfi[set2].describe())
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	50.00000	50.00000	50.00000	50.00000	50.00000
mean	125.50000	6.58800	2.97400	5.55200	2.02600
std	14.57738	0.63588	0.32249	0.55189	0.27465
min	101.00000	4.90000	2.20000	4.50000	1.40000
25%	113.25000	6.22500	2.80000	5.10000	1.80000
50%	125.50000	6.50000	3.00000	5.55000	2.00000
75%	137.75000	6.90000	3.17500	5.87500	2.30000
max	150.00000	7.90000	3.80000	6.90000	2.50000

```
In [35]: set3=(dfi['Species']=='Iris-versicolor')
print(dfi[set3].describe())
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	50.00000	50.000000	50.000000	50.000000	50.000000
mean	75.50000	5.936000	2.770000	4.260000	1.326000
std	14.57738	0.516171	0.313798	0.469911	0.197753
min	51.00000	4.900000	2.000000	3.000000	1.000000
25%	63.25000	5.600000	2.525000	4.000000	1.200000
50%	75.50000	5.900000	2.800000	4.350000	1.300000
75%	87.75000	6.300000	3.000000	4.600000	1.500000
max	100.00000	7.000000	3.400000	5.100000	1.800000

```
In [36]: dfi['Species'].unique()
```

```
Out[36]: array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)
```

```
In [37]: dfi.groupby(dfi['Species']).mean()
```

```
Out[37]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Species					
Iris-setosa	25.5	5.006	3.418	1.464	0.244
Iris-versicolor	75.5	5.936	2.770	4.260	1.326
Iris-virginica	125.5	6.588	2.974	5.552	2.026

```
In [38]: dfi.groupby(dfi['Species']).median()
```

```
Out[38]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Species					
Iris-setosa	25.5	5.0	3.4	1.50	0.2
Iris-versicolor	75.5	5.9	2.8	4.35	1.3
Iris-virginica	125.5	6.5	3.0	5.55	2.0

```
In [39]: dfi.groupby(dfi['Species']).std()
```

```
Out[39]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Species					
Iris-setosa	14.57738	0.352490	0.381024	0.173511	0.107210
Iris-versicolor	14.57738	0.516171	0.313798	0.469911	0.197753
Iris-virginica	14.57738	0.635880	0.322497	0.551895	0.274650

```
In [ ]:
```